

311 Adoption and City Government Spending

1 Data Compilation

The analysis covers all incorporated US places with a 2020 population of at least 100,000, drawn from the Census Bureau’s Population Estimates Program sub-county file ([U.S. Census Bureau, 2023b](#)), joined to the 2023 Census Gazetteer ([U.S. Census Bureau, 2023a](#)) for geographic attributes. This yields 314 cities across 44 states. The 100,000-person threshold matches the population band covered by [O’Byrne \(2015\)](#), the only systematic academic survey of 311 adoption, which enables external validation of search results against a fully independent source.

The primary data-collection method is a structured web search for each of the 314 cities, conducted by a Claude AI agent following a documented protocol. Sources searched in order: the city’s official 311 webpage, mayor’s office announcements, GovTech and Governing magazine coverage, and targeted query strings. Each city produced a structured log file recording the adoption year, source URL, source type (official, news, or secondary), and a one-sentence evidence note. In result, 121 returned a verified adoption year, 55 showed no evidence of a 311 system, and 138 had evidence of 311 but no verifiable launch year.

Search results are cross-validated against [O’Byrne \(2015\)](#), providing a fully independent quality check: the corrected agreement rate is 85.9%, the false-negative rate is zero, and 13 boundary conflicts are excluded to a held-out review set. A formal-311 definitional filter further requires that evidence explicitly reference “311” branding, yielding a final main sample of 301 cities (99 with a verified adoption year, 148 with confirmed 311 but unknown year, 54 with no 311 system).

2 Descriptive Analysis

Figure 1 shows annual and cumulative documented 311 adoptions for the 99 cities with verified launch years. Adoption was gradual rather than concentrated in a single wave: Baltimore adopted in 1996, Chicago and Houston followed in 1999, and uptake continued through the 2000s and 2010s. The figure has two limits. First, 148 additional cities are confirmed 311 adopters but lack verifiable launch years, so the chart understates total prevalence. Second, recent years may be overstated because 2026 web searches more easily recover current launch announcements, creating possible ascertainment bias.

Figure 2 plots within-region adoption shares across the 301 cities in the main sample. The Northeast has the highest share of cities with a documented formal-311 year (52% of 27 cities), while the West contributes 124 cities (mostly California) but the lowest documented-year share (23%). California cities frequently operate 311-adjacent platforms under proprietary branding without accessible launch records, placing the majority in the “adopted, year unknown” category. Across all four regions, the unknown-year segment is substantial,

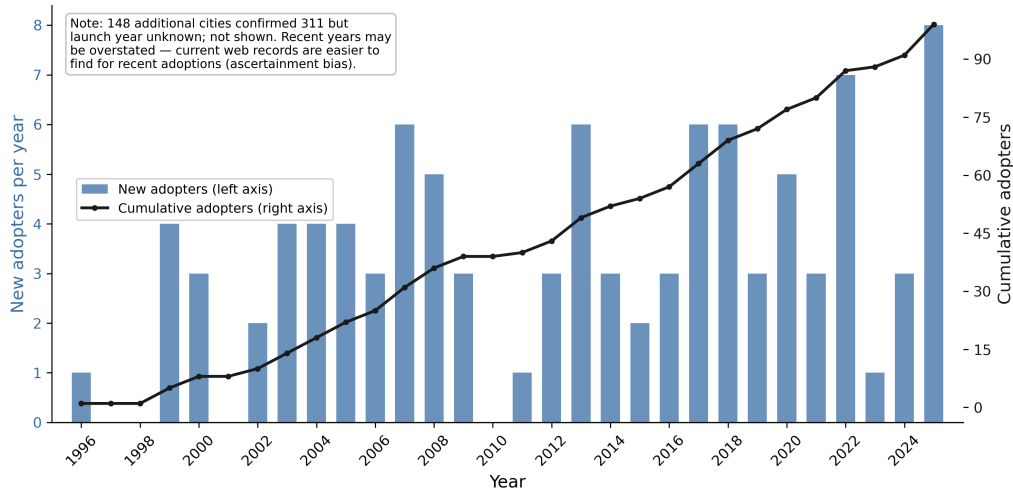


Figure 1: Timing of formal 311 adoption

indicating that the 99-city timing sample undercounts total 311 prevalence and skews toward cities with stronger documentation practices.

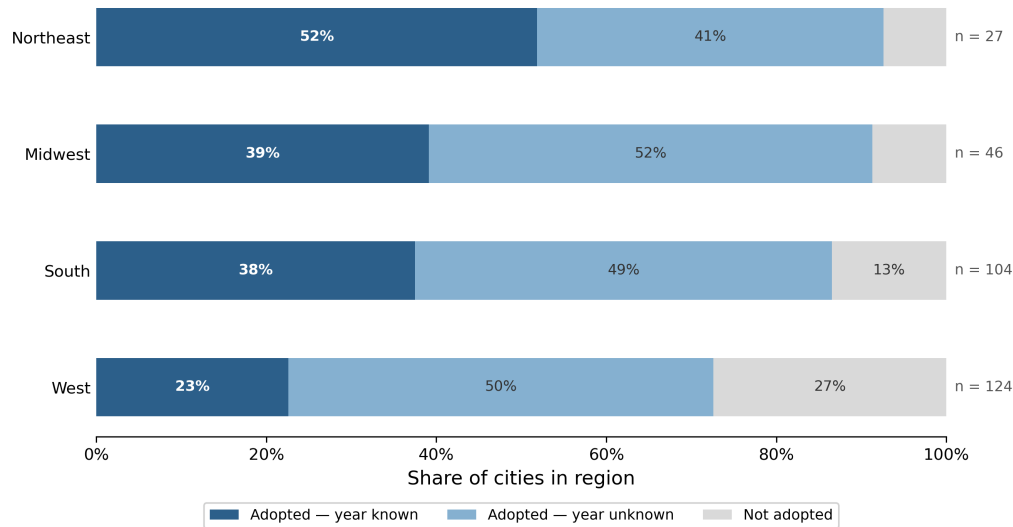


Figure 2: 311 adoption status by Census region, 301 cities. Bars show within-region shares across three adoption statuses. City counts per region are shown at right.

Table 1 compares cities across adoption-status groups. Cities with documented formal-311 adoption years are substantially larger (mean population 497k) than both unknown-year adopters (176k) and non-adopters (236k). This size gradient has two implications. First, the timing analysis in Figure 1 reflects large-city diffusion trajectories, not the full distribution of cities above 100,000 residents. Second, the regression sample in Section 3 skews toward better-documented, larger cities; any fiscal associations estimated there apply primarily to that subpopulation.

Table 1: City characteristics by 311 adoption status. Population in thousands. Regional shares are within-group: the percentage of each group’s cities located in each Census region.

	Not adopted	Adopted (year known)	Adopted (year unknown)	All cities
N	54	99	148	301
Mean population (000s)	236	497	176	292
Median population (000s)	150	211	138	155
Northeast (%)	3.7%	14.1%	7.4%	9.0%
Midwest (%)	7.4%	18.2%	16.2%	15.3%
South (%)	25.9%	39.4%	34.5%	34.6%
West (%)	63.0%	28.3%	41.9%	41.2%

3 Hypothesis Test

To test whether 311 adoption is associated with changes in city government spending, adoption dates from Section 1 are combined with the Census Annual Survey of Government Finances, Individual Unit Data, for 2017 through 2022 (U.S. Census Bureau, 2022). Total general expenditure per capita is constructed from the item-level fixed-width files by summing current-operations (E-prefix), capital-construction (F-prefix), and other-capital-outlay (G-prefix) items, plus general interest on debt (I89), while excluding utility enterprise operations. Cities are matched to the finance data directly via Census place FIPS code, achieving a 100 % match for the 153-city regression sample (99 adopted with known year plus 54 never-adopted cities). The estimator is two-way fixed effects (city and year fixed effects), with standard errors clustered at the city level. The primary data constraint is panel length: Individual Unit Finance files are publicly available only for 2017 through 2022, so cities that adopted before 2017 appear as always-treated throughout the panel with no observable pre-period. The 148 cities with confirmed 311 but unknown adoption timing are excluded from the regression because the post-treatment indicator requires a specific year.

Table 2 reports estimates from two specifications. In the full sample (867 city-year observations), the coefficient on `post311` is -0.025 ($SE = 0.034$, $p = 0.47$), indicating no statistically or economically significant association between 311 adoption and log per capita total general expenditure. A sensitivity specification restricting to cities adopting after 2012 yields a similar estimate of -0.020 ($SE = 0.035$, $p = 0.56$). Both specifications are statistically insignificant; the high table R^2 reflects variance absorbed by city and year fixed effects, not the treatment indicator.

Figure 3 presents event-study estimates for the subset of cities with observable pre- and post-adoption windows in the 2017–2022 panel. The treated group consists of 18 cities that adopted in 2019 through 2022, which are the only cohorts with non-reference pre-period observations given the panel start date. The control group is the 54 never-adopted cities. The event window is $[-3, +3]$ relative to the adoption year. Pre-period coefficients at $k = -3$ and $k = -2$ are small and statistically insignificant (0.047, $SE = 0.060$; 0.012, $SE = 0.033$), consistent with parallel pre-trends. Post-adoption coefficients are also insignificant throughout, though confidence intervals widen as cohort support thins. The bottom panel of the figure reports the number of treated cities contributing to each event-time cell; the coefficient

Table 2: Effect of 311 adoption on log per capita total general expenditure.

	Log(exp. per capita)	
	(1)	(2)
post311	-0.025 (0.034)	-0.020 (0.035)
Year FE	✓	✓
City FE	✓	✓
Observations	867	630
R^2	0.966	0.95

Standard errors clustered at city level. Outcome: log total general expenditure per capita (Census Individual Unit Finance). Col. 1: all treated cities (adopted 1996–2022) + never-adopted controls. Col. 2: late adopters only (adopted 2012–2022) + never-adopted controls. Data: 2017–2022 only; cities adopting before 2017 are always-treated (no pre-period).

at $k = +3$ (three cities) should not be interpreted.

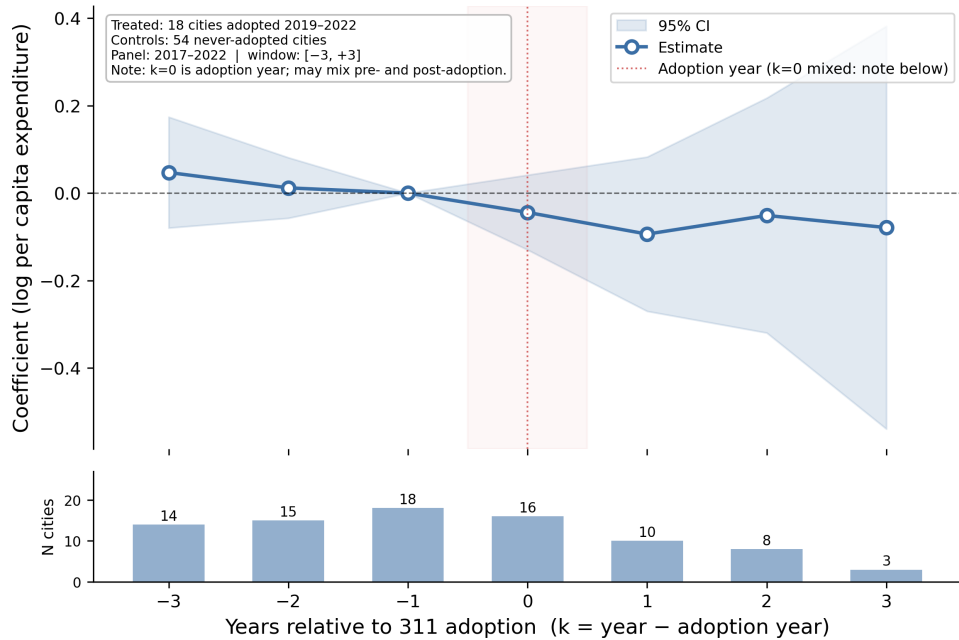


Figure 3: Event-study estimates around 311 adoption.

The null result is informative rather than the product of insufficient power alone. 311 is primarily an administrative routing and case-management system whose direct operating costs are modest relative to total city expenditure. Three limitations qualify the findings. First, the panel covers 2017–2022 only; any fiscal adjustment predating 2017 is undetectable. Second, the event study uses 18 treated cities and is underpowered for detecting moderate-sized effects. Third, the regression sample skews toward larger, better-documented cities

(mean population 497k), which may differ from smaller jurisdictions in how 311 interacts with local fiscal management.

AI tools. Claude assisted throughout: data collection used a Claude agent for structured web searches, and analysis code was developed with Claude Code. The workflow involved iterative human–AI collaboration: planning, AI-assisted execution, and author review and revision at each stage.

References

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